NORTH CAROLINA AGRICULTURAL AND TECHNICAL STATE UNIVERSITY

Department of Electrical and Computer Engineering
(336) 285-3707

SENIOR DESIGN PROJECT
PARTS REQUEST FORM

PART NAME – DESCRIPTION (indicate where in your block diagram your part will be used)	ORDER CONTACT INFORMATION (Company/URL/Contact Name/ph. #)	UNIT PRICE	SUB TOTAL
Pi 5 Cooler (CV Block)	https://www.amazon.com/Official-Active-Cooler-Compatible-Raspberry/dp/B0CW164TCW?crid=3G6FPMPL6J4W&dib=eyJ2IjoiMSJ9.bgg2nbzGXe9bsijcPHqwfSQ3T7CrxoGWDisVNdwg3ctoaAOsruFpnUDeupxYbxxPNHZ2ni2FHLG7Xmx-czTrOD_p9ZIgoHOYN6JTBaXnYv0d8XVj4cvGPxvjeznzCBac0Y0wFb6-plwXlq6CN1PeW6xxBx0MALScn80LiE6IFibM5AkLf9-QhwGCyB-d4OASf-IYvp5Qfauy3_eU5jx7v6uljTxTmBliOqBLyY7wlgg.IqqNuOnTUKhUU9f2TzJS0aPCfRQqZY8-Cofb1XO8hHo&dib_tag=se&keywords=pi%2B5%2Bactive%2Bcooler&qid=1788453694&sprefix=pi%2B5%2Bac%2Caps%2C196&sr=8-3&th=1	$10	$10
IMX219 Camera (Perception Block)	https://www.amazon.com/Arducam-Raspberry-Camera-Module-Megapixel/dp/B083BHJZ16?crid=TWUC3YBHF9G7&dib=eyJ2IjoiMSJ9.er4irRpZg3u2AFO5YgE3Z-Kj1kV3f5qxEhZx8-smEQmsywy4xCE294rdjC7-ueUAjMWcsfBUhljVaGG1SedWWzyR-_ceP6FMIMGz3vBqqxa59RCKZhoG78iY7CLFsfolguKwSo8UFG3puacIXrDRR1lwQ6p8SgMW4oqwCeZAcbBuNjO4zpZl7M0t-HnHW38CXRAjtkxDwyzFWvHoJIttPyvIhWaQR2BXx1YwyAbyD2g.yjl986voCQ-9eTBpBaiGCnHZozvdutQ4lUiUWSg-i-Q&dib_tag=se&keywords=imx219&qid=1788459795&sprefix=Imx%2Caps%2C139&sr=8-4&th=1	$17	$17
Robot Car Kit (Drivetrain Block)	https://www.amazon.com/OSOYOO-FlexiRover-Robot-Raspberry-High-Capacity/dp/B0DJXXH8LC?crid=6O65KBA6KKRR&dib=eyJ2IjoiMSJ9.wng9wPJsE3tqiF8BONElTObmGesahVXSefPbG_kwRL8.6waaEESptmFtg62OdaIOsRFXekRvJAT5GqRsaxrVyiI&dib_tag=se&keywords=osoyoo+flexirover&qid=1788460603&sprefix=flexirove%2Caps%2C216&sr=8-4	$119	$119
Motor w/ encoder qty: 2 (Drivetrain Block)	https://osoyoo.store/products/gear-motor-high-torque-electric-micro-speed-reduction-geared-motor-centric-output-shaft-with-encode	$14.61	$29.22
DRV8871 Breakout qty: 4 (Drivetrain Block)	https://www.adafruit.com/product/3190	$7.50	$30

VL53L0X ToF module (Perception Block)	https://www.amazon.com/ACEIRMC-VL53L0X-Distance-Measurement-Breakout/dp/B0F28MFW6X?crid=2ENNIJOH6NO5G&dib=eyJ2IjoiMSJ9.mt2neMEiDHu3alHNS1n_YcrNkbJMXHul1UFwYcnFXgMyllCpHl5H1tPqWsgp1onV_CkaC-YeATY3-9tpMzIIZqNqFEoZQthPHONRmNyTqLo6X85uMWQRb6f7kiELOVmqajLWqOqGuEmz2DHXTkKF8DSyjOEy_6RI7_Vhb3bk5wvvDwZS6SjrEu8q-z4uKSuMGWy1JPend8q-bFn3Uflxfrf8VaqoqPQ0Wmo4LUH4LUY.EGHmxBEvAcvxYebdcVqPJRRPzLWbmOvPgP4wuOOdZDM&dib_tag=se&keywords=vl53l0x&qid=1788462292&sprefix=vl53l0x%2Caps%2C138&sr=8-4&th=1	$14	$14
IMU LSM6DSOX (Perception Block)	https://www.adafruit.com/product/4438	$12	$12
IO Pin Expander MCP23008 (Perception Block)	https://www.adafruit.com/product/593	$2	$2
Buck Converter (Power Block)	https://www.amazon.com/Converter-DROK-Regulator-Inverter-Transformer/dp/B01NALDSJ0?crid=1ZJT182J9I5KW&dib=eyJ2IjoiMSJ9.zN7WRKX5jauiGXek2ayz6AH64-X6Oqz9mRZDIz9UORMeEOMeRMDitwp4VIBCGH-GEkKdmesPc-x5dFFILtQ43Iyt8hAOi5U_gIsmAJAxh7Q5nI2W4yzx2Gd9lVYGsLKpvlCAIMa7d3xAhYImMOzQwwGH5ujOQ-ZAqZz0NSIG0vFta5H2FMxJM2MqOQUj6RuQlpAIx-U-ihztH4YCr9hVZSozhUwDRWgKUFWJok_8cN0.Lks362ofXb6DpsYo5ieUrsribBumee2yr3RkSzxERJM&dib_tag=se&keywords=buck%2Bconverter%2B12v%2Bto%2B5v%2B5a&qid=1788463376&sprefix=buck%2Bconverter%2B12v%2Bto%2B5v%2B5a%2Caps%2C129&sr=8-6&th=1	$8.70	$8.70
4 Bay Charger (Power Block)	https://www.amazon.com/Battery-Charger-Charge-Rechargeable-Lithium/dp/B094C6GDT2?crid=1O2C2JYAUTYNM&dib=eyJ2IjoiMSJ9._rKAiNQX6atCT6eQw_Qz0uCBKDKtYqy5MI5BGdRBxY3290eJqKCe9rrwHiLn0gBm_VrI2I-xRpGh5eJB2dlHCvvXE1eYKm8C2BwgTuem514dtQfUlz3nXQJd_abLD8T9RVNB9PRGmdGC90GzLkb2Jeklx1q4NxMjcKixO-Q8_uYCs34nU2vhxAdk3wOnFic47tvJueKa1x9sNCjfAK1FLcatA3O7vxBr6LGJkfY3GXY.7eWV_7lK_iG6gOcfiWZX7m6hq-hkhqRlGKr-7qe6huk&dib_tag=se&keywords=4-bay%2B1A%2Bcharger&qid=1788479607&sprefix=4-bay%2B1a%2Bcharger%2Caps%2C221&sr=8-5&th=1	$20	$20
JST-XH 2 pin pigtails (Power Block)	https://www.amazon.com/JST-XH-Connector-Silicone-Battery-Terminal/dp/B0DCNZRJPR?crid=1NLFT9X1PDM50&dib=eyJ2IjoiMSJ9.S-JuN1VXcRHU3AzZ5vVr8dPfPka9NdQIJtg889FSnXURXe3K6JcpABydUovjbULqIt7BP6XBjXDgOpXkRMsAxCfOf4Z0WDZSjU-QETdJWYaBkjgZsvBr4uquz0Qux0P1SNvVOKKF8iGBMwK-DLYyPsxG3eH9Sgu_Izl4ZBvjsXZXxXVd_ucC0D5FQnkLTc0DfolzKAGXiDbQ8zK_xd7lEEcoAkCmtsyobJbVBeC11F4.uxroh0vfWzJ9Lcgxjig7f8zHvB15Jj-I_Hy9puxPtyk&dib_tag=se&keywords=JST-XH+2-pin+pigtails&qid=1788479408&sprefix=jst-xh+2-pin+pigtails%2Caps%2C209&sr=8-4	$7.69	$7.69
Batteries qty: 4 (Power Block)	https://www.18650batterystore.com/products/samsung-30q	$7	$28

Emergency Stop Switch (Power Block)	https://www.amazon.com/STARELO-Mushroom-Emergency-Latching-Normally/dp/B09YD2PNC5?crid=196OV0E6QAQX7&dib=eyJ2IjoiMSJ9.EeU_oPSEV9uP4DJ7F2lz730yGzoz719z27cEiay5EOsrgnNX26Nb2k5Op3PIDXsa6y6KMcYWddcbRXDo5XHU4NmMUPT1x9_3NB26X_T4H3OaRH4OW0XpelYJm5tsVs3E2XdQU2dgNUV7EwvhxbOzyBIEFuFxeHiEnyjC7ZmNgGytL9fnFjidz0XpPNgI_icAeIlMHPqbXbWYI1cUEZJfj1sXmEyKh8RQzIoFd9Hazh7-fZkjtyJ4A3XNjx02UjRXiKF0yJQX5Yr3NP2_IfSPKrNTAFVZLaEZnG_mIbBm59A.HhSLaFho8_VoJ7dv34Sm5Z2Yan3_hksCGgNZEFCfG0I&dib_tag=se&keywords=22mm%2Bmushroom%2BE-stop%2C%2Blatching%2C%2B1NC%2Bcontact%2C%2B10%2BA.&nsdOptOutParam=true&qid=1788480443&s=industrial&sprefix=22mm%2Bmushroom%2Be-stop%2C%2Blatching%2C%2B1nc%2Bcontact%2C%2B10%2Ba.%2Cindustrial%2C135&sr=1-1&th=1	$9	$9
14 gauge Fuse + Holder pack (Power Block)	https://www.amazon.com/SIM-NAT-Automotive-Standard-Replacement/dp/B07FQCBSJ5	$12.49	$12.49
SD Card for Pi (CV Block)	https://www.amazon.com/TAIMI-Micro-UHS-I-Microsd-Memory/dp/B0FXFZC893/ref=sr_1_4?crid=96LXTDX0X1L2&dib=eyJ2IjoiMSJ9.kFT0TNnVjTe6FTPjKNEkC_Mu2py9So393X8G2a2Rz75oR9DNGpUzwIIvrmwo4O7AVKOW8kaskP5p2YDKUUXj1yZ2EG_A-vGfJht8DmquJjuzLtbjcC505uSC6ZoWxF2lE6-sgHrBrJ2xFqgdICIWtCSHmhMj4UHB5jA7AyVjTR88eMchLr_dZP8AYD40hzLZwEzlR4gdMvkTcLt_odanb4y5BzAjKO0L8nh2urdT328.7wkJElc9VaMPgCBsWz7vmHRth5kxNRiJ4r2tpW1s_Ek&dib_tag=se&keywords=microsd%2B32gb&qid=1788836558&refinements=p_36%3A-1400&rnid=386442011&sprefix=microsd%2B32gb%2Caps%2C137&sr=8-4&th=1	$12	$12
22 AWG Wire kit (component interconnects)	https://www.amazon.com/TUOFENG-Solid-Colors-Tinned-Copper/dp/B07TX6BX47/ref=sr_1_5?crid=MVCH3X6LSTGC&dib=eyJ2IjoiMSJ9.PmYOXqiL_EQ6dER1IwGEF7j-hNG2FZO2Ae7L00292pxQuj_IRcVT4gtegnYLc4WLNWCbVgIxpb8NB5LKLUzMv9xQyYb4CusMkU8kddpu3JNuROL7PxfjZdVs8OhVrjxNMry2jJGdDUXMkpfhgwKlur3V3ZS7gftMYz4X7OOQZ4Vx9lbnQFPsWIZh-a6skmoG1-E57ryNdCrv7lLsV-9nC-hmSPl8WIT0PGdU0hkm2CIIHNGcwKTNZTOCjT0hkXCZBGWTxajDIqUfezKZhMDuQM9QWXU0IeksNIo-3XJmUds.Tsz0UP32oIVkFe0UOSGHRpjbMHyiZN7I6P8hIgimwxw&dib_tag=se&keywords=22%2Bawg%2Bwire&qid=1788836728&sprefix=22%2Bawg%2Caps%2C235&sr=8-5&th=1	$16	$16
16 AWG for Motor/Battery connections (Power Block interconnects)	https://www.amazon.com/Voltage-Automotive-Primary-Security-Electrical/dp/B0CZ73SQP8/ref=sr_1_9?crid=AZ6BQ7NAXK7Q&dib=eyJ2IjoiMSJ9.RJwGNkdIKIrQe6E2PMgIWfN8RyE69VyTQzjTCrqbcOMBWKOQKG9rE-pG54HahPr79z7dGC22KFsyYDLGwKRx7m4680PLTJgPBR-RPpIx32LJqok6Piytnpr2K3ZwscrF8yXzjQk14dhknCRndRI4tLk7sudSzNE7qRu1NG40qAE2hEGC75UuoSV9nHUjNBf08aMYYbgIRB82-oN8O-9jSQQ7OMZGkwmCacrSY8VW9PsLYP7CVjeNzuorhU45dfuLSi_KsyYmELMpwqXmwOOf3vlU-EgAwyCDLiwmDUEy_MU.sOuiWK3BUMtjJWhRHDcgqCA4mPd0cCL8WUg7XSj8wX4&dib_tag=se&keywords=16%2Bawg&qid=1788837118&sprefix=16%2Bawg%2Caps%2C156&sr=8-9&th=1	$9	$9
Heat Shrink Tubing (Power Block)	https://www.amazon.com/Rindion-127PCS-Heat-Shrink-Tubing/dp/B0DYSV5Z9B/ref=sr_1_8?crid=266BL3XULPXPO&dib=eyJ2IjoiMSJ9.SXkRC94fgf4LOuGQ9aeLWS8l9nMvBmyGJ9YYtPecc8Cl3c8Sxl9tswZExGIUpAFFNHNXZM4FezMqjY7NUZsMmKWzNBQNHoboxtgHdIw8PsNO0zI0RWeGaCbAYuvqrse0Ew2YKozMybZq_bosq70MBt010SUfYA727ZcJsLSCbfareY2ejU50-R9l-T1dyrjEZOZlL-5BzxhiSY-00BP4m2jJo9VhVQXY6-3BsBH8fHY.1r-kIc68yuUZYgW-ALfPWiZRCgMhWY8jDJgxlnUynVw&dib_tag=se&keywords=heat%2Bshrink&qid=1788837262&sprefix=heat%2Bshrink%2Caps%2C168&sr=8-8&th=1	$5	$5

Nylon Standoffs (component mounting; all blocks)	https://www.amazon.com/PATIKIL-Motherboard-Standoffs-Assortment-Electrical/dp/B0DGPZXRCW/ref=sr_1_14?crid=DJWCBZ34QB4J&dib=eyJ2IjoiMSJ9.gdQ5h6Ta8LxIpM4YtBDE1CAgL4_4XbZsyfJlbCDVwz0gOH-MRL483N80EQnW0PeTG3iJdSwFX9i1ET3mhoZ0k2W3gcicqkiuU2cXQIsAy7umiP-zyEpZ-ubPfgbKlfkOPXBIipmMR1sjZGILIFDdwcNmtz46DPITlD0vaA4n8tJmKZ41KE87Cn8orsQ2UXqto9nI1K0aRVfLexlD3H5pBxnflMnoSIJOipYsZJ1TnNY.F9SWgv_nAPRXwJAN3KthQACbZmpfR6qfE-zwIJC9YMA&dib_tag=se&keywords=M2.5+nylon+standoff&qid=1788837403&sprefix=m2.5+nylon+standoff%2Caps%2C161&sr=8-14	$10	$10
Perfboard Kit (component assembly; all blocks)	https://www.amazon.com/WWZMDiB-Circuit-Protoboard-Soldering-Arduino/dp/B0CMZL4T6J?crid=B5YTO81JJYZS&dib=eyJ2IjoiMSJ9.jKh2ezKT6h9Gz7qSfYOh5vxWM9bN3mB_8bd6bf0zvPeAYS8fiMepjsX24L2cz8P1a2UiC7LMB3dWYzgO1gZB38uavLKdJ_Mfal8QqlS9RXBMUlTGRErEusO-38D5pZI2ZVO29X_pyZbdIsBlDoKZqwk2LM5HUeu3Afy6WNv2w_wrAm-dg1kUU1Pi1q5szsAjawb44SQIGKuX--LKQ3exNYeqfmdFqqLb74113ckKZ0U.XyiYYwcJ6XHt6gaSHxeRSPuWvgToTDfe0H7jeHaTyXE&dib_tag=se&keywords=perfboard+kit&qid=1789058705&sprefix=perfbord%2Caps%2C142&sr=8-11	$10	$10
LiDAR Sensor (Perception block)	https://www.amazon.com/DTOF-D300-Distance-Obstacle-Education/dp/B0B1V8D36H?crid=YXC2CVXPQAU1&dib=eyJ2IjoiMSJ9.2FRSibZD4woaYjEGAaBCEYyI9R95lHmrzT1E4pME6NBq1MJHvReWpvIMhYko-QWTq05gMm9XcI5kfENdWSuVSIuBIhO-yuOVQP5clu5t4E8MKB3n6L0HV3wadsPynaP5Ox9Mg-r54LL2z1bdyEX5e34AjLSLBCn77yCmucWHft-xm1Wt9pLb6Km_KHTQWTSbcD9dJGbk5AIucYsKZJDG-jRKobAA03-5DR2xFJUa9VE.QHLkRVo2Wm6VYPbezYXGtdAnG_qfzJSdhKlelWyGOe4&dib_tag=se&keywords=lidar+sensor&qid=1789057711&sprefix=lidar%2Caps%2C107&sr=8-3	$70	$70
TOTAL			$451

****Please attach a copy of your most up-to-date block diagram**

GROUP NAME:_____

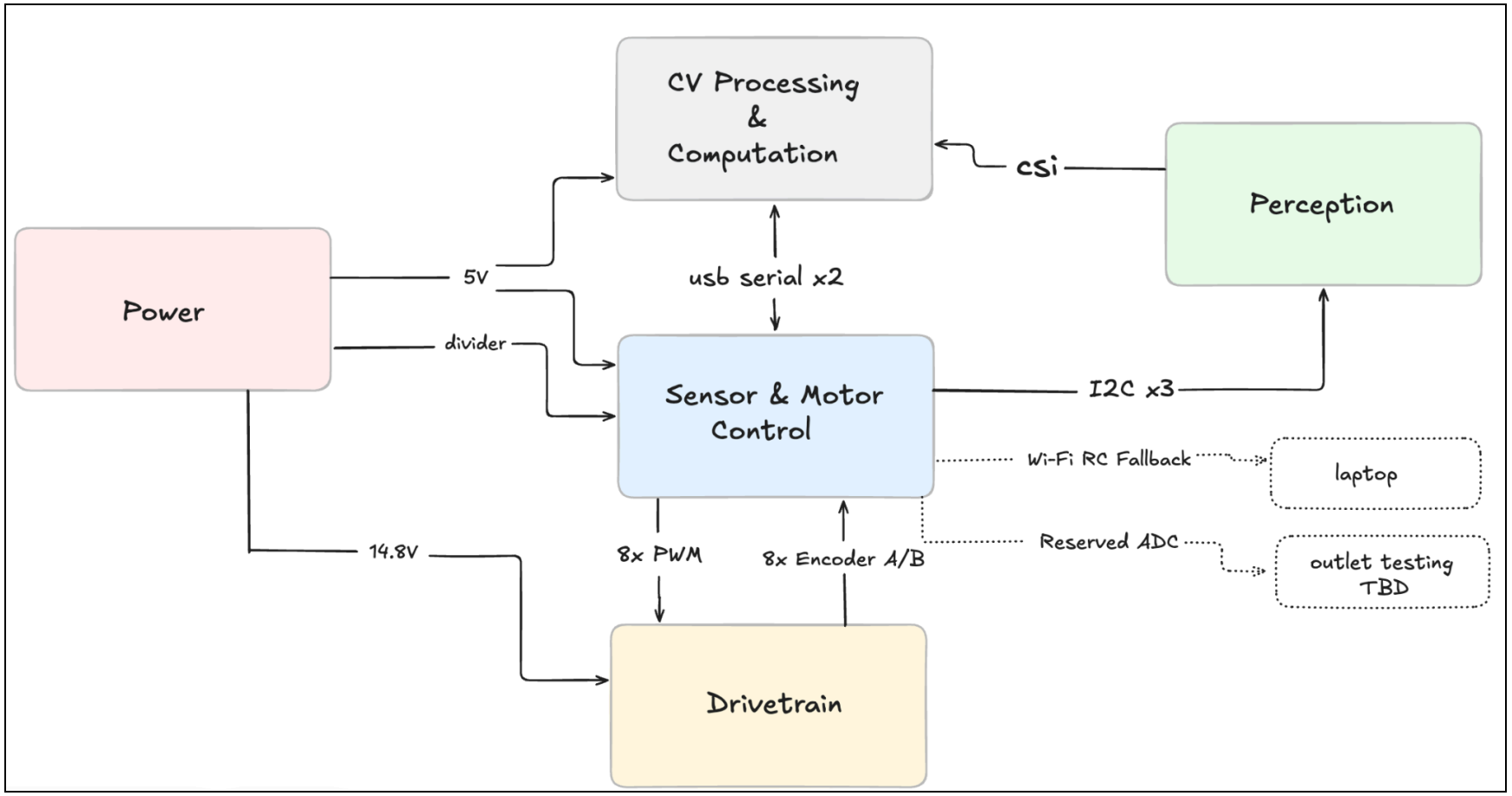
GROUP LEADER _____ EMAIL ADDRESS: _____

PROJECT TITLE: _____ DATE: _____

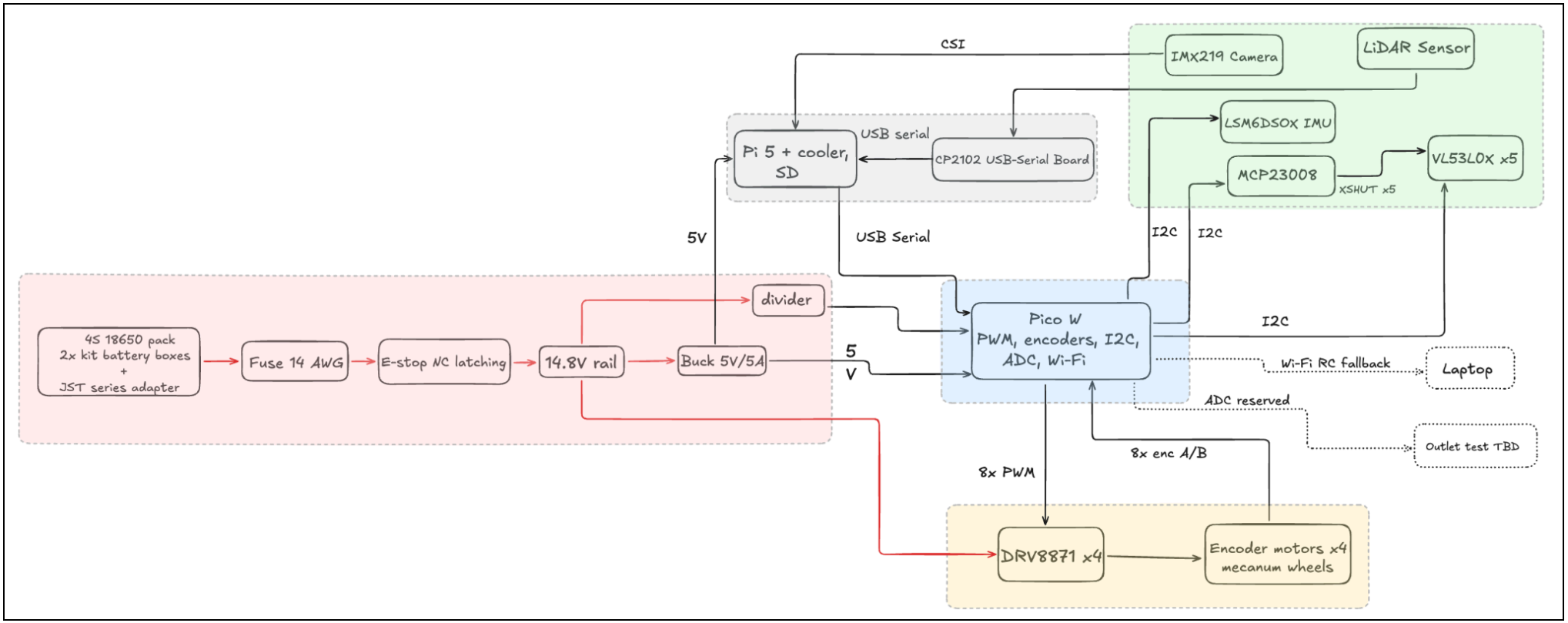
FACULTY ADVISER: _____ SIGNATURE:_____DATE:_____

CLASS INSTRUCTOR: Dr. Abul-Fadl SIGNATURE:_____DATE: _____

DATE PROCESSED:_____



Top Level Diagram: Five functional blocks. Power conditions the 4S lithium-ion pack into the rails the system needs. Perception collects visual, ranging, and inertial data. CV Processing & Computation (Raspberry Pi 5) handles outlet detection, mapping, and navigation planning. Sensor & Motor Control (Pico W) handles the real-time layer, motor commands and sensor polling. Drivetrain converts commands into wheel motion and returns encoder feedback.



Component Level Diagram:
Each block expanded into its physical parts.

Wiring: red paths are the unregulated battery rail in 16 AWG (fuse holder leads are 14 AWG). Black paths are signal or low-current lines in 22 AWG.

Four 30Q cells in series form a 14.8V pack, routed through an inline fuse and a normally-closed latching E-stop that cuts all downstream power. The rail feeds the motor drivers directly and a 5V / 5A buck converter supplying the Pi 5, Pico W, lidar, and sensor breakouts. A resistor divider taps the rail ahead of the buck so the Pico can monitor pack voltage for under-voltage cutoff. The Pi 5 takes the IMX219 camera over CSI and the LD19 lidar over USB serial, and talks to the Pico over USB serial. The Pico runs one I2C bus carrying the IMU, an MCP23008, and the VL53L0X sensors, the expander drives their XSHUT lines so each can be readdressed at boot, since they share a default address. The Pico also generates 8 PWM channels into four DRV8871 drivers and reads 8 encoder channels back from the four mecanum-wheel motors.